Untitled

October 24, 2026

[1]:
```
import pandas as pd
```

[2]:
```
pd.__version__
```

[2]:
```
'3.0.4'
```

[3]:
```
df = pd.read_csv("onlinefoods.csv")
```

[4]:
```
df
```

[4]:

	Age	Gender	Marital Status	Occupation	Monthly Income	\
0	20	Female	Single	Student	No Income	
1	24	Female	Single	Student	Below Rs.10000	
2	22	Male	Single	Student	Below Rs.10000	
3	22	Female	Single	Student	No Income	
4	22	Male	Single	Student	Below Rs.10000	
..	…	…	…	…	…	
383	23	Female	Single	Student	No Income	
384	23	Female	Single	Student	No Income	
385	22	Female	Single	Student	No Income	
386	23	Male	Single	Student	Below Rs.10000	
387	23	Male	Single	Student	No Income	

	Educational Qualifications	Family size	latitude	longitude	Pin code	\
0	Post Graduate	4	12.9766	77.5993	560001	
1	Graduate	3	12.9770	77.5773	560009	
2	Post Graduate	3	12.9551	77.6593	560017	
3	Graduate	6	12.9473	77.5616	560019	
4	Post Graduate	4	12.9850	77.5533	560010	
..	…	…	…	…	…	
383	Post Graduate	2	12.9766	77.5993	560001	
384	Post Graduate	4	12.9854	77.7081	560048	
385	Post Graduate	5	12.9850	77.5533	560010	
386	Post Graduate	2	12.9770	77.5773	560009	
387	Post Graduate	5	12.8988	77.5764	560078	

	Output	Feedback	Unnamed: 12
0	Yes	Positive	Yes

```
1      Yes   Positive         Yes
2      Yes  Negative          Yes
3      Yes   Positive         Yes
4      Yes   Positive         Yes
..     …        …          …
383    Yes   Positive         Yes
384    Yes   Positive         Yes
385    Yes   Positive         Yes
386    Yes   Positive         Yes
387    Yes   Positive         Yes

[388 rows x 13 columns]
```

[5]:
```
df.head(5)
```

[5]:
```
   Age  Gender Marital Status Occupation  Monthly Income  \
0   20  Female         Single    Student       No Income
1   24  Female         Single    Student  Below Rs.10000
2   22    Male         Single    Student  Below Rs.10000
3   22  Female         Single    Student       No Income
4   22    Male         Single    Student  Below Rs.10000

  Educational Qualifications  Family size  latitude  longitude  Pin code  \
0              Post Graduate            4   12.9766    77.5993    560001
1                   Graduate            3   12.9770    77.5773    560009
2              Post Graduate            3   12.9551    77.6593    560017
3                   Graduate            6   12.9473    77.5616    560019
4              Post Graduate            4   12.9850    77.5533    560010

  Output   Feedback Unnamed: 12
0    Yes   Positive         Yes
1    Yes   Positive         Yes
2    Yes  Negative          Yes
3    Yes   Positive         Yes
4    Yes   Positive         Yes
```

[6]:
```
df.tail(5)
```

[6]:
```
     Age  Gender Marital Status Occupation  Monthly Income  \
383   23  Female         Single    Student       No Income
384   23  Female         Single    Student       No Income
385   22  Female         Single    Student       No Income
386   23    Male         Single    Student  Below Rs.10000
387   23    Male         Single    Student       No Income

    Educational Qualifications  Family size  latitude  longitude  Pin code  \
383              Post Graduate            2   12.9766    77.5993    560001
```

```
384              Post Graduate            4   12.9854    77.7081    560048
385              Post Graduate            5   12.9850    77.5533    560010
386              Post Graduate            2   12.9770    77.5773    560009
387              Post Graduate            5   12.8988    77.5764    560078

    Output  Feedback Unnamed: 12
383    Yes  Positive         Yes
384    Yes  Positive         Yes
385    Yes  Positive         Yes
386    Yes  Positive         Yes
387    Yes  Positive         Yes
```

[7]:
```
df.describe
```

[7]:
```
<bound method NDFrame.describe of      Age  Gender Marital Status Occupation
Monthly Income  \
0     20  Female         Single    Student       No Income
1     24  Female         Single    Student  Below Rs.10000
2     22    Male         Single    Student  Below Rs.10000
3     22  Female         Single    Student       No Income
4     22    Male         Single    Student  Below Rs.10000
..   …     …             …         …              …
383   23  Female         Single    Student       No Income
384   23  Female         Single    Student       No Income
385   22  Female         Single    Student       No Income
386   23    Male         Single    Student  Below Rs.10000
387   23    Male         Single    Student       No Income

    Educational Qualifications  Family size  latitude  longitude  Pin code  \
0                Post Graduate            4   12.9766    77.5993    560001
1                     Graduate            3   12.9770    77.5773    560009
2                Post Graduate            3   12.9551    77.6593    560017
3                     Graduate            6   12.9473    77.5616    560019
4                Post Graduate            4   12.9850    77.5533    560010
..                         …          …       …        …       …
383              Post Graduate            2   12.9766    77.5993    560001
384              Post Graduate            4   12.9854    77.7081    560048
385              Post Graduate            5   12.9850    77.5533    560010
386              Post Graduate            2   12.9770    77.5773    560009
387              Post Graduate            5   12.8988    77.5764    560078

    Output   Feedback Unnamed: 12
0      Yes   Positive         Yes
1      Yes   Positive         Yes
2      Yes  Negative          Yes
3      Yes   Positive         Yes
4      Yes   Positive         Yes
```

```
..     …        …          …
383    Yes   Positive        Yes
384    Yes   Positive        Yes
385    Yes   Positive        Yes
386    Yes   Positive        Yes
387    Yes   Positive        Yes

[388 rows x 13 columns]>
```

[8]:
```
df.dtypes
```

[8]:
```
Age                             int64
Gender                            str
Marital Status                    str
Occupation                        str
Monthly Income                    str
Educational Qualifications        str
Family size                     int64
latitude                      float64
longitude                     float64
Pin code                        int64
Output                            str
Feedback                          str
Unnamed: 12                       str
dtype: object
```

[9]:
```
type(df["latitude"][1])
```

[9]:
```
numpy.float64
```

[10]:
```
df.shape
```

[10]:
```
(388, 13)
```

[11]:
```
 df.info
```

[11]:
```
<bound method DataFrame.info of      Age  Gender Marital Status Occupation
Monthly Income  \
0     20  Female         Single    Student       No Income
1     24  Female         Single    Student  Below Rs.10000
2     22    Male         Single    Student  Below Rs.10000
3     22  Female         Single    Student       No Income
4     22    Male         Single    Student  Below Rs.10000
..   …     …             …        …              …
383   23  Female         Single    Student       No Income
384   23  Female         Single    Student       No Income
385   22  Female         Single    Student       No Income
386   23    Male         Single    Student  Below Rs.10000
```

```
387   23    Male         Single    Student       No Income

    Educational Qualifications  Family size  latitude  longitude  Pin code  \
0                Post Graduate            4   12.9766    77.5993    560001
1                     Graduate            3   12.9770    77.5773    560009
2                Post Graduate            3   12.9551    77.6593    560017
3                     Graduate            6   12.9473    77.5616    560019
4                Post Graduate            4   12.9850    77.5533    560010
..                         …          …       …        …        …
383              Post Graduate            2   12.9766    77.5993    560001
384              Post Graduate            4   12.9854    77.7081    560048
385              Post Graduate            5   12.9850    77.5533    560010
386              Post Graduate            2   12.9770    77.5773    560009
387              Post Graduate            5   12.8988    77.5764    560078

    Output   Feedback Unnamed: 12
0      Yes   Positive         Yes
1      Yes   Positive         Yes
2      Yes  Negative          Yes
3      Yes   Positive         Yes
4      Yes   Positive         Yes
..     …        …          …
383    Yes   Positive         Yes
384    Yes   Positive         Yes
385    Yes   Positive         Yes
386    Yes   Positive         Yes
387    Yes   Positive         Yes

[388 rows x 13 columns]>
```

[12]:
```
df.columns
```

[12]:
```
Index(['Age', 'Gender', 'Marital Status', 'Occupation', 'Monthly Income',
       'Educational Qualifications', 'Family size', 'latitude', 'longitude',
       'Pin code', 'Output', 'Feedback', 'Unnamed: 12'],
      dtype='str')
```

[13]:
```
df.columns
```

[13]:
```
Index(['Age', 'Gender', 'Marital Status', 'Occupation', 'Monthly Income',
       'Educational Qualifications', 'Family size', 'latitude', 'longitude',
       'Pin code', 'Output', 'Feedback', 'Unnamed: 12'],
      dtype='str')
```

[14]:
```
columns_to_keep = ['Age', 'Gender', 'Marital Status', 'Occupation', 'Monthly␣
↪Income',
       'Educational Qualifications', 'Family size', 'Output', 'Feedback']
```

```
columns_to_drop = ['latitude', 'longitude',
       'Pin code', 'Unnamed: 12']
```

[15]:
```
len(columns_to_keep)
```

[15]: 9

[16]:
```
len(columns_to_drop)
```

[16]: 4

[17]:
```
#Filterring columns what we need
```

[18]:
```
df
```

[18]:

	Age	Gender	Marital Status	Occupation	Monthly Income	\
0	20	Female	Single	Student	No Income	
1	24	Female	Single	Student	Below Rs.10000	
2	22	Male	Single	Student	Below Rs.10000	
3	22	Female	Single	Student	No Income	
4	22	Male	Single	Student	Below Rs.10000	
..	…	…	…	…	…	
383	23	Female	Single	Student	No Income	
384	23	Female	Single	Student	No Income	
385	22	Female	Single	Student	No Income	
386	23	Male	Single	Student	Below Rs.10000	
387	23	Male	Single	Student	No Income	

	Educational Qualifications	Family size	latitude	longitude	Pin code	\
0	Post Graduate	4	12.9766	77.5993	560001	
1	Graduate	3	12.9770	77.5773	560009	
2	Post Graduate	3	12.9551	77.6593	560017	
3	Graduate	6	12.9473	77.5616	560019	
4	Post Graduate	4	12.9850	77.5533	560010	
..	…	…	…	…	…	
383	Post Graduate	2	12.9766	77.5993	560001	
384	Post Graduate	4	12.9854	77.7081	560048	
385	Post Graduate	5	12.9850	77.5533	560010	
386	Post Graduate	2	12.9770	77.5773	560009	
387	Post Graduate	5	12.8988	77.5764	560078	

	Output	Feedback	Unnamed: 12
0	Yes	Positive	Yes
1	Yes	Positive	Yes
2	Yes	Negative	Yes
3	Yes	Positive	Yes
4	Yes	Positive	Yes

```
..     …        …          …
383    Yes   Positive         Yes
384    Yes   Positive         Yes
385    Yes   Positive         Yes
386    Yes   Positive         Yes
387    Yes   Positive         Yes

[388 rows x 13 columns]
```

[19]:
```
df[columns_to_keep]
```

[19]:
```
     Age  Gender Marital Status Occupation  Monthly Income  \
0     20  Female         Single    Student       No Income
1     24  Female         Single    Student  Below Rs.10000
2     22    Male         Single    Student  Below Rs.10000
3     22  Female         Single    Student       No Income
4     22    Male         Single    Student  Below Rs.10000
..   …     …             …         …               …
383   23  Female         Single    Student       No Income
384   23  Female         Single    Student       No Income
385   22  Female         Single    Student       No Income
386   23    Male         Single    Student  Below Rs.10000
387   23    Male         Single    Student       No Income

    Educational Qualifications  Family size Output   Feedback
0                Post Graduate            4    Yes   Positive
1                     Graduate            3    Yes   Positive
2                Post Graduate            3    Yes  Negative
3                     Graduate            6    Yes   Positive
4                Post Graduate            4    Yes   Positive
..                         …          …    …        …
383              Post Graduate            2    Yes   Positive
384              Post Graduate            4    Yes   Positive
385              Post Graduate            5    Yes   Positive
386              Post Graduate            2    Yes   Positive
387              Post Graduate            5    Yes   Positive

[388 rows x 9 columns]
```

[20]:
```
modified_data = df[columns_to_keep]
```

[21]:
```
modified_data
```

[21]:
```
     Age  Gender Marital Status Occupation  Monthly Income  \
0     20  Female         Single    Student       No Income
1     24  Female         Single    Student  Below Rs.10000
2     22    Male         Single    Student  Below Rs.10000
```

```
3     22  Female         Single    Student       No Income
4     22    Male         Single    Student  Below Rs.10000
..   ...     ...            ...        ...              ...
383   23  Female         Single    Student       No Income
384   23  Female         Single    Student       No Income
385   22  Female         Single    Student       No Income
386   23    Male         Single    Student  Below Rs.10000
387   23    Male         Single    Student       No Income

    Educational Qualifications  Family size Output   Feedback
0                Post Graduate            4    Yes   Positive
1                     Graduate            3    Yes   Positive
2                Post Graduate            3    Yes  Negative
3                     Graduate            6    Yes   Positive
4                Post Graduate            4    Yes   Positive
..                         ...          ...    ...        ...
383              Post Graduate            2    Yes   Positive
384              Post Graduate            4    Yes   Positive
385              Post Graduate            5    Yes   Positive
386              Post Graduate            2    Yes   Positive
387              Post Graduate            5    Yes   Positive

[388 rows x 9 columns]
```

[22]:
```
#Dropping Unnecessary columns
```

[23]:
```
df.drop(columns = columns_to_drop,inplace = True)
```

[24]:
```
df
```

[24]:
```
     Age  Gender Marital Status Occupation  Monthly Income  \
0     20  Female         Single    Student       No Income
1     24  Female         Single    Student  Below Rs.10000
2     22    Male         Single    Student  Below Rs.10000
3     22  Female         Single    Student       No Income
4     22    Male         Single    Student  Below Rs.10000
..   ...     ...            ...        ...              ...
383   23  Female         Single    Student       No Income
384   23  Female         Single    Student       No Income
385   22  Female         Single    Student       No Income
386   23    Male         Single    Student  Below Rs.10000
387   23    Male         Single    Student       No Income

    Educational Qualifications  Family size Output   Feedback
0                Post Graduate            4    Yes   Positive
1                     Graduate            3    Yes   Positive
2                Post Graduate            3    Yes  Negative
```

```
3                    Graduate            6    Yes   Positive
4                Post Graduate           4    Yes   Positive
..                         …           …    …       …
383              Post Graduate           2    Yes   Positive
384              Post Graduate           4    Yes   Positive
385              Post Graduate           5    Yes   Positive
386              Post Graduate           2    Yes   Positive
387              Post Graduate           5    Yes   Positive

[388 rows x 9 columns]
```

[25]:
```
df.columns
```

[25]:
```
Index(['Age', 'Gender', 'Marital Status', 'Occupation', 'Monthly Income',
       'Educational Qualifications', 'Family size', 'Output', 'Feedback'],
      dtype='str')
```

[26]:
```
df.rename(columns = {"Monthly Income": "Monthly Salary"},inplace = True)
```

[27]:
```
df
```

[27]:
```
     Age  Gender Marital Status Occupation  Monthly Salary  \
0     20  Female         Single    Student       No Income
1     24  Female         Single    Student  Below Rs.10000
2     22    Male         Single    Student  Below Rs.10000
3     22  Female         Single    Student       No Income
4     22    Male         Single    Student  Below Rs.10000
..   …     …             …         …              …
383   23  Female         Single    Student       No Income
384   23  Female         Single    Student       No Income
385   22  Female         Single    Student       No Income
386   23    Male         Single    Student  Below Rs.10000
387   23    Male         Single    Student       No Income

    Educational Qualifications  Family size Output   Feedback
0                Post Graduate           4    Yes   Positive
1                     Graduate           3    Yes   Positive
2                Post Graduate           3    Yes  Negative
3                     Graduate           6    Yes   Positive
4                Post Graduate           4    Yes   Positive
..                         …           …    …       …
383              Post Graduate           2    Yes   Positive
384              Post Graduate           4    Yes   Positive
385              Post Graduate           5    Yes   Positive
386              Post Graduate           2    Yes   Positive
387              Post Graduate           5    Yes   Positive
```

```
[388 rows x 9 columns]
```

[28]:
```
df
```

[28]:
```
     Age  Gender Marital Status Occupation  Monthly Salary  \
0     20  Female         Single    Student       No Income
1     24  Female         Single    Student  Below Rs.10000
2     22    Male         Single    Student  Below Rs.10000
3     22  Female         Single    Student       No Income
4     22    Male         Single    Student  Below Rs.10000
..   ...     ...            ...        ...              ...
383   23  Female         Single    Student       No Income
384   23  Female         Single    Student       No Income
385   22  Female         Single    Student       No Income
386   23    Male         Single    Student  Below Rs.10000
387   23    Male         Single    Student       No Income

    Educational Qualifications  Family size Output   Feedback
0                Post Graduate            4    Yes   Positive
1                     Graduate            3    Yes   Positive
2                Post Graduate            3    Yes  Negative
3                     Graduate            6    Yes   Positive
4                Post Graduate            4    Yes   Positive
..                         ...          ...    ...        ...
383              Post Graduate            2    Yes   Positive
384              Post Graduate            4    Yes   Positive
385              Post Graduate            5    Yes   Positive
386              Post Graduate            2    Yes   Positive
387              Post Graduate            5    Yes   Positive

[388 rows x 9 columns]
```

[29]:
```
df
```

[29]:
```
     Age  Gender Marital Status Occupation  Monthly Salary  \
0     20  Female         Single    Student       No Income
1     24  Female         Single    Student  Below Rs.10000
2     22    Male         Single    Student  Below Rs.10000
3     22  Female         Single    Student       No Income
4     22    Male         Single    Student  Below Rs.10000
..   ...     ...            ...        ...              ...
383   23  Female         Single    Student       No Income
384   23  Female         Single    Student       No Income
385   22  Female         Single    Student       No Income
386   23    Male         Single    Student  Below Rs.10000
387   23    Male         Single    Student       No Income
```

```
    Educational Qualifications  Family size Output   Feedback
0                Post Graduate            4    Yes   Positive
1                     Graduate            3    Yes   Positive
2                Post Graduate            3    Yes  Negative
3                     Graduate            6    Yes   Positive
4                Post Graduate            4    Yes   Positive
..                         …          …    …        …
383              Post Graduate            2    Yes   Positive
384              Post Graduate            4    Yes   Positive
385              Post Graduate            5    Yes   Positive
386              Post Graduate            2    Yes   Positive
387              Post Graduate            5    Yes   Positive

[388 rows x 9 columns]
```

[30]:
```
df.duplicated().sum()
```

[30]:
```
np.int64(144)
```

[31]:
```
df.drop_duplicates(inplace= True)
```

[32]:
```
df
```

[32]:
```
     Age  Gender Marital Status Occupation   Monthly Salary  \
0     20  Female         Single    Student        No Income
1     24  Female         Single    Student   Below Rs.10000
2     22    Male         Single    Student   Below Rs.10000
3     22  Female         Single    Student        No Income
4     22    Male         Single    Student   Below Rs.10000
..   …     …             …         …               …
355   21    Male         Single    Student        No Income
363   31    Male        Married   Employee  More than 50000
369   30    Male        Married   Employee  More than 50000
374   21    Male         Single    Student        No Income
386   23    Male         Single    Student   Below Rs.10000

    Educational Qualifications  Family size Output   Feedback
0                Post Graduate            4    Yes   Positive
1                     Graduate            3    Yes   Positive
2                Post Graduate            3    Yes  Negative
3                     Graduate            6    Yes   Positive
4                Post Graduate            4    Yes   Positive
..                         …          …    …        …
355                   Graduate            2     No   Positive
363                       Ph.D            3    Yes   Positive
369              Post Graduate            6    Yes   Positive
374                   Graduate            3    Yes  Negative
```

```
386              Post Graduate                2    Yes    Positive

[244 rows x 9 columns]
```

[33]:
```
df.isna().sum()
```

[33]:
```
Age                           0
Gender                        0
Marital Status                0
Occupation                    0
Monthly Salary                0
Educational Qualifications    0
Family size                   0
Output                        0
Feedback                      0
dtype: int64
```

[34]:
```
df.dropna(inplace =True)
```

[35]:
```
df.isna().sum()
```

[35]:
```
Age                           0
Gender                        0
Marital Status                0
Occupation                    0
Monthly Salary                0
Educational Qualifications    0
Family size                   0
Output                        0
Feedback                      0
dtype: int64
```

[36]:
```
df["Age (Months)"] = df["Age"] * 12
```

[37]:
```
df
```

[37]:
```
     Age  Gender Marital Status Occupation   Monthly Salary  \
0     20  Female         Single    Student        No Income
1     24  Female         Single    Student   Below Rs.10000
2     22    Male         Single    Student   Below Rs.10000
3     22  Female         Single    Student        No Income
4     22    Male         Single    Student   Below Rs.10000
..   …     …            …        …               …
355   21    Male         Single    Student        No Income
363   31    Male        Married   Employee  More than 50000
369   30    Male        Married   Employee  More than 50000
374   21    Male         Single    Student        No Income
386   23    Male         Single    Student   Below Rs.10000
```

```
    Educational Qualifications  Family size Output   Feedback  Age (Months)
0                Post Graduate            4    Yes   Positive           240
1                     Graduate            3    Yes   Positive           288
2                Post Graduate            3    Yes  Negative            264
3                     Graduate            6    Yes   Positive           264
4                Post Graduate            4    Yes   Positive           264
..                         …          …    …       …            …
355                   Graduate            2     No   Positive           252
363                       Ph.D            3    Yes   Positive           372
369              Post Graduate            6    Yes   Positive           360
374                   Graduate            3    Yes  Negative            252
386              Post Graduate            2    Yes   Positive           276

[244 rows x 10 columns]
```

[38]:
```
df["Gender"] = df["Gender"].apply(lambda x: "F" if x == "Female" else "M")
```

[39]:
```
df.head(5)
```

[39]:
```
   Age Gender Marital Status Occupation  Monthly Salary  \
0   20      F          Single    Student       No Income
1   24      F          Single    Student  Below Rs.10000
2   22      M          Single    Student  Below Rs.10000
3   22      F          Single    Student       No Income
4   22      M          Single    Student  Below Rs.10000

  Educational Qualifications  Family size Output   Feedback  Age (Months)
0              Post Graduate            4    Yes   Positive           240
1                   Graduate            3    Yes   Positive           288
2              Post Graduate            3    Yes  Negative            264
3                   Graduate            6    Yes   Positive           264
4              Post Graduate            4    Yes   Positive           264
```

[40]:
```
# if family size > 3 it's going to be large else small
```

[41]:
```
df["Family size"] = df["Family size"].apply(lambda x: "Large" if x > 3 else␣
↪"Small")
```

[42]:
```
df.head(2)
```

[42]:
```
   Age Gender Marital Status Occupation  Monthly Salary  \
0   20      F          Single    Student       No Income
1   24      F          Single    Student  Below Rs.10000

  Educational Qualifications Family size Output  Feedback  Age (Months)
0              Post Graduate       Large    Yes  Positive           240
```

```
1                   Graduate       Small    Yes  Positive           288
```

[43]:
```
df["Output"] = df["Output"].apply(lambda x: 1 if x == "Yes" else 0)
```

[44]:
```
df.head(5)
```

[44]:
```
   Age Gender Marital Status Occupation  Monthly Salary  \
0   20      F          Single    Student       No Income
1   24      F          Single    Student  Below Rs.10000
2   22      M          Single    Student  Below Rs.10000
3   22      F          Single    Student       No Income
4   22      M          Single    Student  Below Rs.10000

  Educational Qualifications Family size  Output   Feedback  Age (Months)
0              Post Graduate       Large       1   Positive           240
1                   Graduate       Small       1   Positive           288
2              Post Graduate       Small       1  Negative            264
3                   Graduate       Large       1   Positive           264
4              Post Graduate       Large       1   Positive           264
```

[45]:
```
df["Output"] = df["Output"].astype(float)
```

[46]:
```
df.head(3)
```

[46]:
```
   Age Gender Marital Status Occupation  Monthly Salary  \
0   20      F          Single    Student       No Income
1   24      F          Single    Student  Below Rs.10000
2   22      M          Single    Student  Below Rs.10000

  Educational Qualifications Family size  Output   Feedback  Age (Months)
0              Post Graduate       Large     1.0   Positive           240
1                   Graduate       Small     1.0   Positive           288
2              Post Graduate       Small     1.0  Negative            264
```

[47]:
```
df.to_excel("Cleaned_Data.xlsx", index=False)
```